

SAN FRANCISCO INTL, CA, USA

WMO: 724940

Lat: 37.620N

Lon: 122.365W

Elev: 8

StdP: 14.69

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 23234

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 40.3	(c) 42.2	(d) 26.2	(e) 20.2	(f) 52.1	(g) 29.8	(h) 23.9	(i) 51.5	(j) 31.4	(k) 56.3	(l) 26.5	(m) 55.3	(n) 3.5	(o) 150	(p) 0.583

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9	16.0	82.9	62.8	78.1	62.0	74.5	61.5	65.9	77.4	64.2	74.4	63.0	72.1	12.6	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.6	81.9	68.9	60.2	77.9	67.5	59.0	74.6	66.3	30.5	77.4	29.3	74.5	28.4	72.0	78.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 28.5	(b) 25.5	(c) 23.3	(e) 36.2	(f) 92.9	(g) 2.1	(h) 3.6	(i) 34.7	(j) 95.5	(k) 33.5	(l) 97.6	(m) 32.3	(n) 99.6	(o) 30.8	(p) 102.2		
(a) 28.5	(b) 25.5	(c) 23.3	(e) 33.3	(f) 69.4	(g) 2.4	(h) 2.4	(i) 31.5	(j) 71.1	(k) 30.1	(l) 72.5	(m) 28.7	(n) 73.8	(o) 27.0	(p) 75.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	58.3	51.3	53.0	55.2	56.8	59.2	62.0	63.3	64.2	64.8	62.2	56.2	51.4
	DBStd	6.29	3.78	3.99	4.17	4.34	4.25	4.22	3.29	3.44	4.65	4.55	4.20	4.11
	HDD50	84	29	14	4	1	0	0	0	0	0	0	5	31
	HDD65	2606	424	336	305	251	191	112	74	56	57	113	266	421
	CDD50	3128	70	98	166	206	287	359	413	441	445	378	190	75
	CDD65	173	0	0	1	5	12	21	22	32	52	27	1	0
	CDH74	945	0	0	10	46	90	154	97	125	278	140	5	0
	CDH80	267	0	0	0	10	26	52	22	34	89	34	0	0
(14) Wind	WSAvg	10.3	6.6	8.3	10.0	12.4	13.5	13.7	12.7	12.2	10.9	9.0	7.0	7.0
(15) Precipitation	PrecAvg	21.9	4.2	4.1	3.0	1.5	0.7	0.2	0.0	0.1	0.1	1.1	2.3	4.6
	PrecMax	38.0	10.4	14.3	8.7	5.0	3.3	1.6	0.1	0.9	0.5	3.7	7.2	11.8
	PrecMin	8.1	0.3	0.0	0.2	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.4
	PrecStd	6.3	3.1	3.0	2.4	1.2	0.8	0.4	0.0	0.2	0.1	1.1	1.8	3.2
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	65.1	70.0	74.6	80.1	83.5	86.8	82.5	85.1	88.9	84.5	73.0	64.5
		MCWB	53.1	56.8	59.0	60.7	61.7	63.6	65.2	64.7	64.4	61.0	59.3	57.3
	2%	DB	61.4	64.6	68.4	72.2	74.7	77.8	76.0	76.6	81.5	77.5	68.0	61.6
		MCWB	54.1	55.9	56.6	57.9	59.6	61.7	63.2	63.7	63.3	59.5	57.3	56.3
	5%	DB	59.1	61.6	64.2	66.6	69.3	72.3	72.3	73.0	76.0	73.0	65.1	59.8
		MCWB	53.5	55.1	55.6	56.2	58.6	60.6	61.8	62.5	62.6	59.2	56.7	55.2
	10%	DB	57.2	59.1	61.5	63.3	65.7	69.1	69.8	70.5	72.1	69.5	62.7	58.0
		MCWB	52.7	53.7	54.7	54.9	56.7	59.1	60.7	61.4	61.8	59.0	56.2	53.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	58.9	60.3	61.6	62.7	65.3	66.2	66.3	66.9	68.4	65.6	62.4	60.8
		MCDB	60.8	64.1	70.1	75.0	77.1	81.8	79.5	78.5	79.4	75.2	68.4	62.7
	2%	WB	56.4	58.0	59.1	59.7	62.0	63.2	63.9	64.6	65.9	63.3	60.4	58.9
		MCDB	58.7	62.2	65.2	68.5	71.4	75.3	74.3	74.3	76.8	71.9	65.1	60.5
	5%	WB	54.9	56.4	57.3	57.7	59.4	61.3	62.4	63.2	64.0	61.8	59.0	56.8
		MCDB	57.3	60.2	62.1	64.8	67.3	71.5	71.4	71.9	73.9	70.0	63.3	58.7
	10%	WB	53.4	54.8	55.8	55.8	57.6	59.6	61.0	61.9	62.3	60.1	57.4	54.6
		MCDB	55.9	58.2	60.3	62.1	64.6	68.2	69.1	70.1	71.1	67.5	61.4	56.9
(35) Mean Daily Temperature Range	5% DB	MDBR	10.6	11.6	13.1	13.8	13.8	15.2	14.6	14.5	16.0	15.4	12.5	10.5
		MCDBR	13.3	15.7	18.2	20.6	21.2	22.2	18.8	18.8	23.9	22.1	16.0	11.5
		MCWBR	8.1	9.1	10.2	10.6	10.3	10.4	8.8	8.5	11.1	10.7	8.8	7.7
	5% WB	MCDBR	9.5	13.1	15.5	18.3	18.9	20.9	18.1	18.2	21.6	19.0	13.1	8.2
		MCWBR	7.1	8.3	9.6	10.3	9.8	10.1	8.7	8.4	10.4	10.0	8.4	7.0
(40) Clear-Sky Solar Irradiance	taub		0.324	0.332	0.341	0.358	0.365	0.359	0.367	0.345	0.348	0.338	0.322	0.324
	taud		2.562	2.537	2.504	2.421	2.426	2.449	2.432	2.494	2.470	2.540	2.568	2.566
	Ebn at Noon		275	286	293	292	290	291	288	293	287	280	273	266
	Edh at Noon		24	28	31	36	36	36	36	33	32	27	23	23
(44) All-Sky Solar Radiation	RadAvg		764	1058	1494	1952	2220	2410	2246	2000	1715	1279	895	667
	RadStd		102	127	129	150	148	105	82	98	69	81	57	95

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	-2.02	N/A	N/A	N/A	N/A	N/A	+245	+40
(47) Regional (44 neighbors)	N/A	N/A	-2.14	N/A	N/A	N/A	N/A	-198	+243	N/A

Nomenclature: See separate page